

Manyu Simha Ravi

Bengaluru, India | manyusimha26@gmail.com | +91 8296479899
github.com/manyu2024 | linkedin | https://manyu2024.github.io(Portfolio site)

Professional Summary

Third-year Computer Science and Cyber Security undergraduate with hands-on experience in offensive security, web application testing, network enumeration, privilege escalation, and packet analysis in lab environments. Proficient in Burp Suite, Nmap, Metasploit, Wireshark, Python, and Linux-based environments. Ranked in the top 25% globally on TryHackMe and actively pursuing the ISC2 Certified in Cybersecurity (CC) certification. Seeking an internship where I can learn how the industry functions in real-world environments, work alongside experienced professionals, and continue building practical offensive security skills.

Education

Dayananda Sagar College of Engineering
Bachelor of Engineering in Computer Science & Cyber Security

Expected June 2027
Bengaluru, India

- Relevant Coursework: Computer Networks, Operating Systems, Linux CLI, Python, Database Management Systems (DBMS), Data Structures and Algorithms, Cyber Security Fundamentals.

Work History

Offensive Security Practitioner — Self-Directed Home Lab

September 2025 – Present

(For detailed writeups, visit <https://manyu2024.github.io>)

- Performed stack-based buffer overflow exploitation to understand memory corruption vulnerabilities.
- Performed DNS zone transfer attacks against intentionally misconfigured BIND servers to demonstrate information disclosure risks.
- Executed ARP poisoning and Man-in-the-Middle (MITM) attacks using Ettercap and Wireshark to analyze network traffic interception techniques.
- Exploited vulnerable services including VSFTPD, UnrealIRCd, and Samba to obtain remote shell access and perform post-exploitation activities.
- Assessed local network devices for misconfigurations and unauthenticated access vulnerabilities.
- Identified an exposed printing endpoint on a network printer that allowed unauthorized print job submission through direct IP access.
- Documented exploitation steps, impact, and remediation notes for each assessment.

Cultural Lead — DSCE Cyber Security Club (Aegis)

January 2025 – Present

- Led a 12-member team in organizing SANDBOX, a national-level cybersecurity hackathon.
- Coordinated multiple inter-departmental technical and cultural events as Cultural Head.
- Developed leadership, public speaking, and team coordination skills through event management and community engagement initiatives.

Projects

FINN — Terminal AI Security Assistant

November 2024 – Present

Tech Stack: Python, Llama 3.3 70B (Groq API), Rich

- Developed a 600+ line terminal-based AI security assistant designed to help with penetration testing workflows, CVE lookups, and security tool references during assessments.
- Integrated Groq-hosted Llama 3.3 70B models to provide contextual cybersecurity assistance directly within a command-line interface.
- Implemented a hybrid JSON-based memory system for persistent user profiles, session context retention, and conversation history management.
- Integrated web search for unknown tools, CVEs, and security concepts.
- Designed a terminal-style interface with command panels, typewriter effects, and live session stats using Rich.

Penetration Testing Lab Environment

March 2026 – Present

Tech Stack: Kali Linux, Metasploitable, VirtualBox

- Designed and maintained an isolated virtual penetration testing environment using Kali Linux, Metasploitable, and VirtualBox.
- Conducted vulnerability assessments and exploitation exercises against intentionally vulnerable targets in a controlled lab setting.
- Leveraged Metasploit Framework and Meterpreter payloads to gain remote shell access, perform privilege escalation, and validate exploitation paths.
- Practiced enumeration, exploitation, post-exploitation, and reporting workflows following standard penetration testing workflow: enumeration, exploitation, post-exploitation, reporting.

Technical Skills

Offensive Security & Penetration Testing: Web Application Security, SQL Injection (SQLi), Cross-Site Scripting (XSS), CSRF, IDOR, Path Traversal, Network Enumeration, Privilege Escalation, MITM Attacks, Buffer Overflow Exploitation, Password Cracking

Security Tools: Burp Suite, Nmap, Metasploit Framework, sqlmap, Hydra, John the Ripper, Hashcat, Gobuster, Dirbuster, Amass, Wireshark, Ettercap, Netcat

Platforms & Labs: Kali Linux, TryHackMe (Top 20% globally), Hack The Box (10+ machines), PortSwigger Web Security Academy, OverTheWire Bandit, DVWA, Metasploitable, VirtualBox

Programming & Scripting: Python, Bash, Git, GitHub

Networking & Operating Systems: TCP/IP, DNS, HTTP/HTTPS, Packet Analysis, Linux CLI, Linux File Permissions, Subnetting

CTF Competitions & Platforms

- Participated in various Capture The Flag (CTF) competitions covering web exploitation, cryptography, OSINT, forensics, and binary exploitation challenges.
- Ranked in the top 25% globally on TryHackMe and completed learning paths focused on offensive security, enumeration, exploitation, and red team methodologies.
- Solved multiple Hack The Box machines involving network enumeration, web application exploitation, privilege escalation, and post-exploitation techniques.
- Completed practical labs on PortSwigger Web Security Academy, OverTheWire Bandit, and DVWA covering common web vulnerabilities and Linux fundamentals.

Certifications

- ISC2 Certified in Cybersecurity (CC) — *In Progress*
- Cisco Networking Academy — Introduction to Cybersecurity
- CySec IISc — Cyber Security MOOC

Achievements

- Participated in the CySec CTF conducted by IISc, focusing on web exploitation, cryptography, and OSINT challenges.
- Secured 2nd Place in Code Sprint (Code Relay Competition) at the DSCE CSD Department Technical Fest.